

Structure of our analysis

- Context and literature review
- Description of our data collection methodology and strategy
- Data analysis and economical implications
- Display of possible extensions and limitations
- Appendix

Do prison commissaries exploit their monopolistic market power?

Evidence from web-scraped prices: comparing Walkenhorst and Californian Walmart Prices

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Context and literature review

Economic context: captive consumers and monopoly provision

*What is the **magnitude of the price differences** for matched products?*

Are observed price differences consistent with monopoly pricing in a captive market?

Do price markups vary across product categories?

Economic interest of the questions:

- Incarcerated individuals face zero consumer choice or substitution, with no exit option
- Prison commissaries operate as monopoly providers

We thus aim at **evaluating the market power and potential markups of a monopoly over a legally captive population** in comparison to the competitive market.

In addition, **important social considerations are at stake** as purchases are most often funded by low-income families, and face prices that directly affect welfare, nutrition, and dignity.

Monopoly vs Competitive Market Prices

California's carceral system: trends and policy relevance

Despite California's reputation as a progressive state, it is **a center of mass incarceration** in the United States of America, detaining more people than any other state besides Texas.

The California State Prison system, administered by **the California Department of Correction and Rehabilitation (CDCR) oversees a population of approximately 90,000 inmates** (as of January 21st, 2026), with an operating budget of 14.3 billion dollars (2024-2025 budget allocation).

In California, incarceration isn't peripheral, it is **large-scale, data-rich matter that directly impacts households**, constituting an ideal setting to study market dynamics.



California's carceral landscape: geographic spread



Map of California's Correctional and Rehabilitation Institutions



- Almost every Californian county has at least one jail, and state-wide there are **31 state prison facilities and approximately 90,000 inmates.**
- California's prison population has been declining substantially over the past years. The state has closed three prisons since 2021 and plans another closure in 2025.

Yet, even with downsizing, the incarceration system remains significant as well as **institutional and private purchasing markets.**

Existing evidence on commissary pricing and inequality

Bardelli, Gillespie & Tu (2022), *“Surviving Austerity: Commissary Stores, Inequality and Punishment in the Contemporary American Prison”* (*Punishment & Society*).

- Depict commissaries as **a mechanism of financial extraction** which burdens incarcerated people and their families.
- Estimate the average yearly commissary purchases of \$1,249 in New York.
- Warn that rising commissary prices combined with stagnant prison wages exacerbate inequality and welfare losses.

Raher (2020), *“The Company Store and the Literally Captive Market: Consumer Law in Prisons and Jails,”*

- States that the **growth of public expense associated with mass incarceration has led many carceral systems to push certain costs onto the people**, over “literally captive market”.
- Describes the consolidation of commissary vendors, on fee-based businesses that are typically outsourced to private firms.
- Discusses potential consumer-law implications and **unfair practices**.

Commissary pricing in the public debate

North Carolina Prison Officials Raise Prices for Those Who Can Least Afford It

An increased markup on canteen goods adds to the financial burden on incarcerated people and their families, highlighting how prison systems squeeze them for revenue.

Phillip Vance Smith II | December 2, 2025

*“North Carolina Prison Officials Raise Prices for Those Who Can Least Afford It”,
Bolts Magazine (Dec 2, 2025)*

Denounces the increase in coverage of official markups on canteen goods for incarcerated workers.

🌟 “Locked In, Priced Out”: Markups and Kickbacks in Prison Commissaries

Loaded on JAN. 15, 2025 published in Prison Legal News January, 2025, page 48

Filed under: Food/Commissary (Private Prisons), Seizure of Prisoner Funds, Commissary. Location: United States of America.

“Locked in, Priced out: Markups and kickbacks in prison”, Prison Legal News (Jan 15, 2025)

Shows commissary groceries and necessities marked up up to five times the community price, with some markups as high as 600%.

KITES

Inflation Doesn't Stop at the Prison Gates

With rising prices and low pay, incarcerated people struggle to meet even their most basic needs.

by AMANDA EWING
December 11, 2025

“Inflation Doesn't Stop at the Prison Gates”, Prison Journalism Project (Dec 11, 2025)

Shows commissary prices rising on essentials goods, putting pressure on incarcerated people and families (First-hand account from Arkansas).

Data collection: methodology and strategy

Data collection process

Correctional authorities require families who wish to send packages to incarcerated people to do so through approved private vendors such as Walkenhorst. **Vendors provide prescreened catalogs,** and ship packages directly to correctional facilities. In the state of California, **Walkenhorst is the sole vendor for all CDCR facilities.**

*This system is designed to reduce contraband and simplify security screening.
It creates a regulated retail market with limited entry and restricted consumer choice.*

2025 FAMILY VISITATION CATALOG			WALKENHORST'S		
Item #	Description	Price	Item #	Description	Price
3111-153	Charleston Chew Chocolate 2.1 oz.	\$1.25	86383-173	Ice Breakers Cinnamon Mints 1.5 oz.	\$2.95
3111-151	Charleston Chew Strawberry 2.1 oz.	\$1.25	86383-172	Ice Breakers DUO Watermelon Mints 1.3 oz.	\$2.95
3111-152	Charleston Chew Vanilla 2.1 oz.	\$1.25	86383-191	Ice Breakers Sparkling Pineapple Mango Seltzer Mints 1.5 oz.	\$2.95
92426-110	Charms Fluffy Stuff Cotton Candy - Original 3.5 oz.	\$2.75	22669-282	Jell-O Pudding Cups 1.55 oz.	\$1.45
22669-237	® Chocolate Lovers Mini Variety Pack 8.9 oz	\$6.95	22669-281	Jell-O Pudding Cups 3.5 oz.	\$2.75
86383-085	® Chocolatey Payday King Size 3.1 oz.	\$2.45	22669-285	Jell-O Sour Candy Squares Berry Blue 4.5 oz.	\$2.45
62267-250	® Chunky 1.4 oz.	\$1.25	86383-300	Jolly Rancher Assorted Fruit Flavors 7 oz.	\$3.25
92426-018	® Cookie Dough Bites Chocolate Chip 3.1 oz.	\$2.25	86383-083	Jolly Rancher Gummies Original 7 oz.	\$2.95
62267-233	® Crunch 1.55 oz.	\$1.25	86383-084	Jolly Rancher Gummies Sours 6.5 oz.	\$2.95
58182-199	Crunch Buncha Crunch 8 oz.	\$6.45	86383-175	Jolly Rancher Gummies Very Berry 6.5 oz.	\$2.95
62267-251	® Crunch Fun Size 10 oz.	\$6.75	86383-176	Jolly Rancher Hard Candy Tropical 6.5 oz.	\$2.95
58421-017	Darrell Lea Rocklea Road Block 6.4 oz	\$4.95	46266-276	Jovy Chili Covered Mango & Watermelon Candy 6 oz.	\$2.50
58182-002	® Dove Dark Chocolate Bar 1.3 oz.	\$1.25	3110-711	® Joyva Kosher Halvah Marble 1.75 oz.	\$1.25
58182-003	® Dove Milk Chocolate Bar 1.44 oz.	\$1.25	92426-149	® Junior Mints 2.6 oz.	\$2.45

Challenges in collecting the support data

Objective:

Construct a harmonized merged dataset with comparable unit prices for identical goods sold in different market environments: over heterogenous sources and products categories for which formats and naming conventions that differ substantially across sources.

- From **Walkenhorst** catalogue, the main challenge was to reliably parse the structured tables into a matching ready data frame (string separation by pattern recognition in R).
- From **Walmart** website, we scraped underlying JSON files by specific category search queries, circumventing **both consequent detection and access limits issues**.
- Scrapped data provided inconsistently formatted data with unit formats (¢/oz, \$/oz, \$/lb, \$/count), including repeated observations of identical products offered in multiple sizes.

Data harmonization and matching strategy

Data harmonization process:

- Standardization of weight units, conversion of prices to a standard unit (dollars per ounce).
- Dropping observations not relevant to our analysis (e.g. Walmart store brand products).
- Text cleaning removing numbers, weights, counts, special characters from product names to reduce noise in the data.

Aggregation and merging of the data with fuzzy matching in Python (RapidFuzz and FuzzyWuzzy libraries):

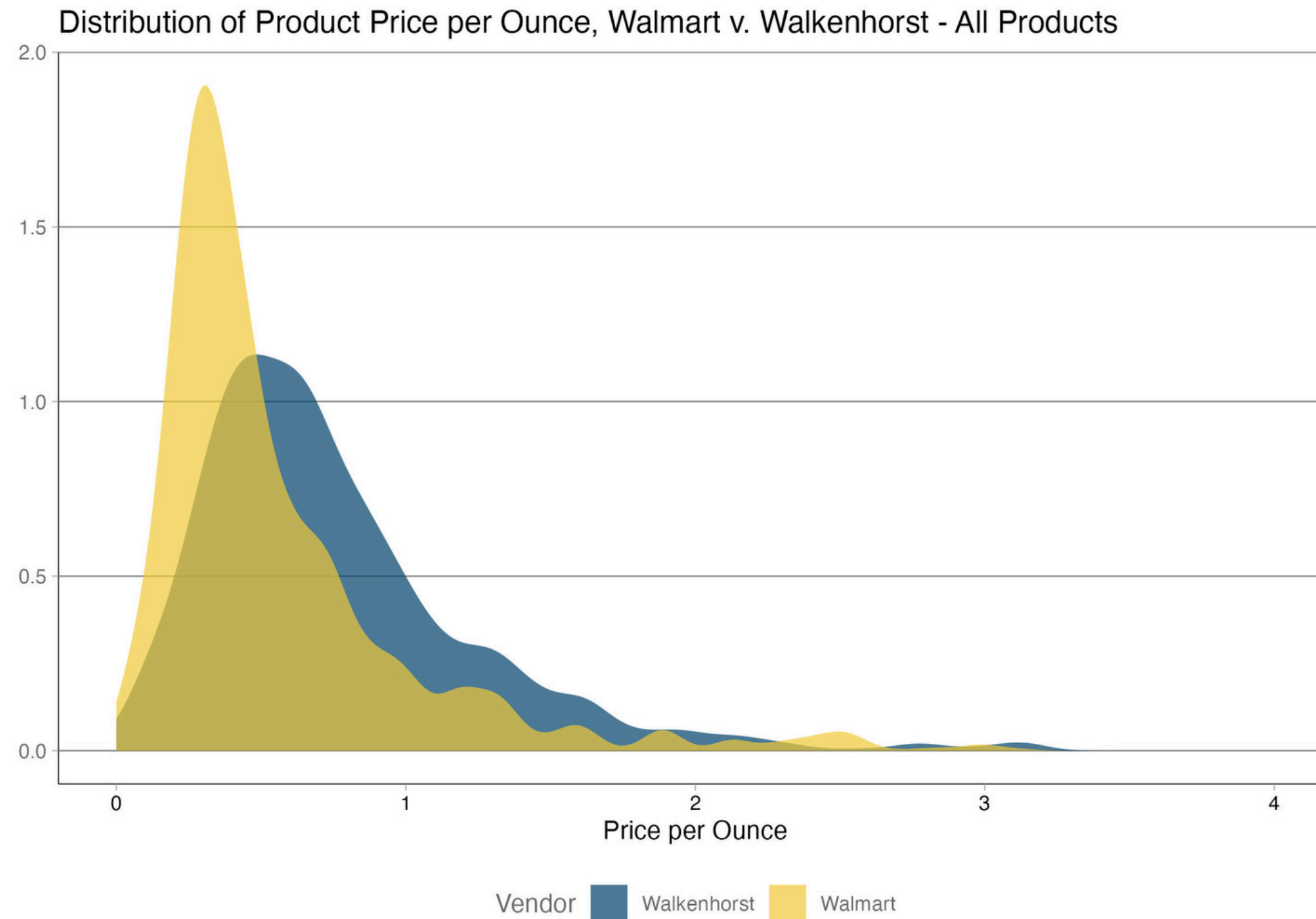
- Using token-based similarity to compare product names and compute match scores.
- Retaining matches with score $\geq 65\%$ to limit the selection of false positives.
- Then, matching products with sufficient confidence scores: prioritizing match accuracy over maximization of observations number.

Our approach thus implies a trade-off:

Balancing sample size and match accuracy

Analysis and public policy recommendations

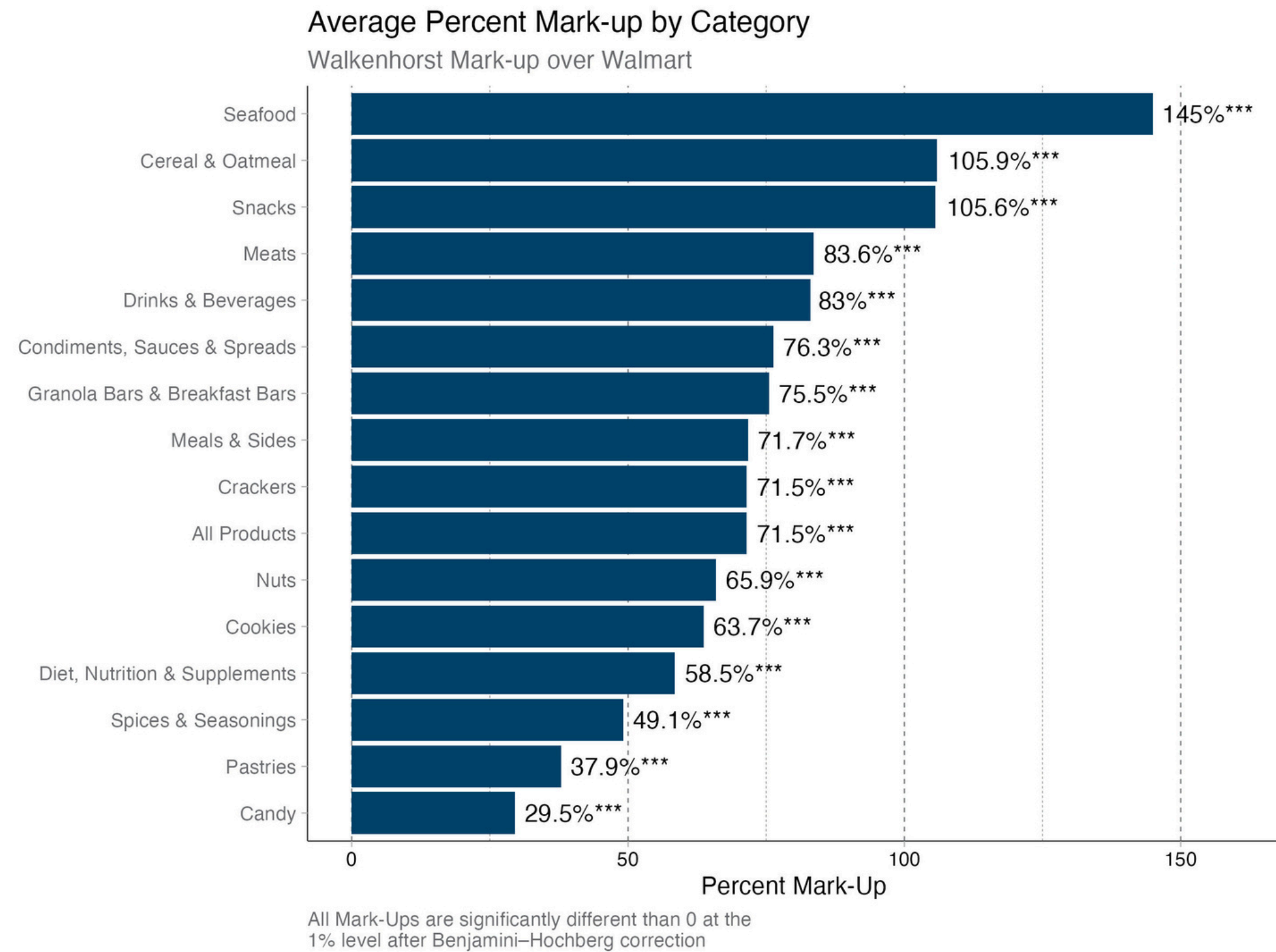
Walmart v. Walkenhorst: unit prices visualised



Price-per-unit difference is significant at the 1% level after Benjamini-Hochberg correction

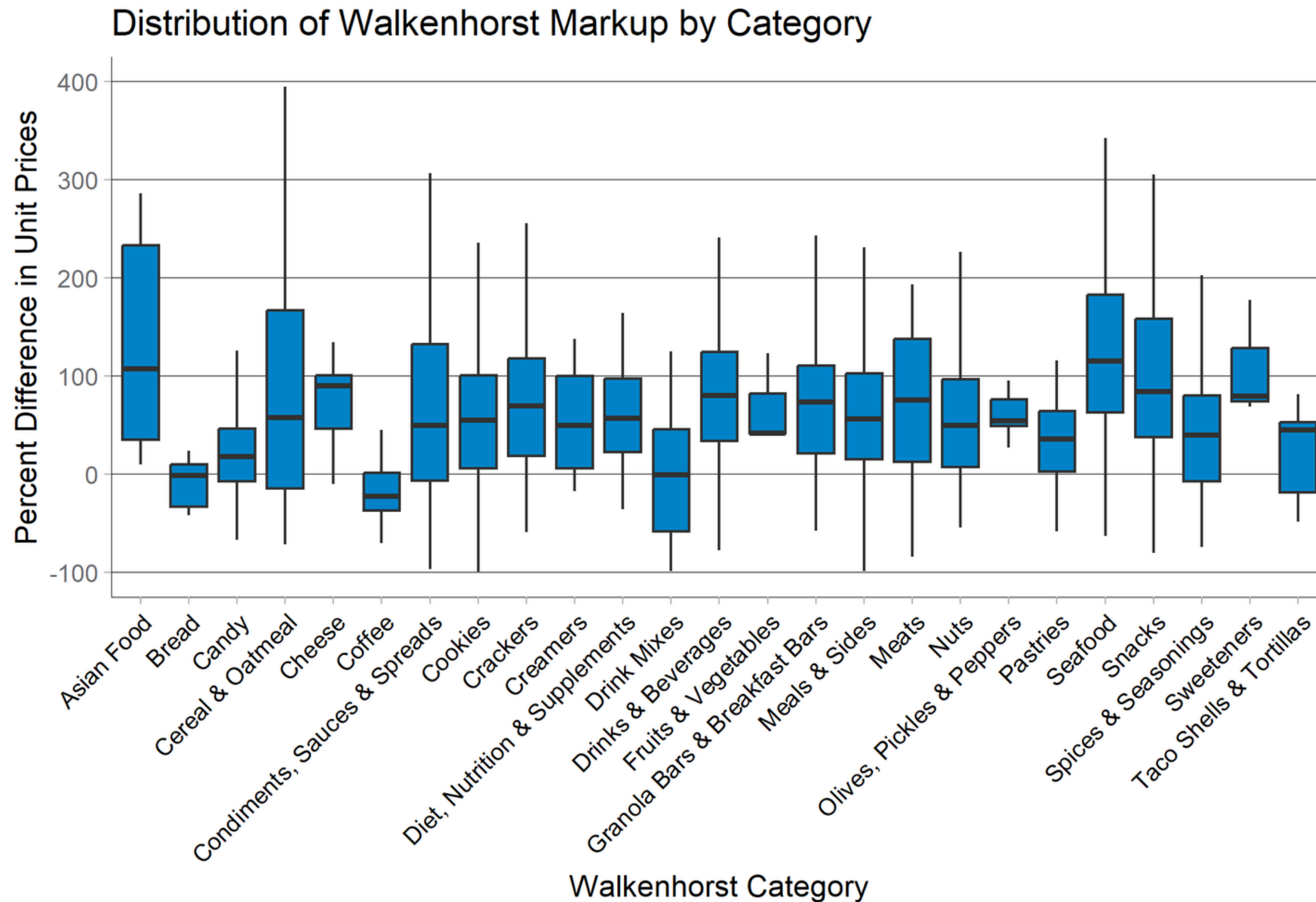
Walmart prices per ounce are more tightly concentrated at lower values, while Walkenhorst exhibits a broader distribution with a heavier right tail, suggesting systematically higher and more dispersed unit prices at Walkenhorst.

Percent markup-visualized



We observe a systematic markup on all products categories, from 29.5% to 145%.
On average, Walkenhorst products are higher by 71.5%

Box plot of markup by category



- Overall, most categories have a positive median markup and a positive IQR distribution, with a few exceptions.
- High degree of variability between categories.
- Some have highly left/right-skewed distributions but most are roughly centered.
- Many categories have relatively clustered distributions of percent markup, but others have a very wide spread.

Markup test results

Table 1: Markup Test Results

Category	Mean Markup (%)	Degrees of Freedom	t-stat	P-Value
SEAFOOD	144.99***	61.00	9.63	0.00
CEREAL & OATMEAL	105.89***	45.00	4.27	0.00
SNACKS	105.59***	201.00	15.36	0.00
MEATS	83.62***	34.00	4.86	0.00
DRINKS & BEVERAGES	82.99***	30.00	5.59	0.00
CONDIMENTS, SAUCES & SPREADS	76.32***	113.00	6.36	0.00
GRANOLA BARS & BREAKFAST BARS	75.50***	58.00	8.14	0.00
MEALS & SIDES	71.67***	199.00	10.68	0.00
CRACKERS	71.51***	33.00	5.84	0.00
All Products	71.49***	1399.00	27.03	0.00
NUTS	65.94***	50.00	5.45	0.00
COOKIES	63.71***	114.00	7.24	0.00
DIET, NUTRITION & SUPPLEMENTS	58.52***	51.00	7.35	0.00
SPICES & SEASONINGS	49.11***	84.00	5.35	0.00
PASTRIES	37.89***	69.00	4.76	0.00
CANDY	29.54***	133.00	4.82	0.00

Note: * $p<0.1$; ** $p<0.05$; *** $p<0.01$.

$H0: \text{markup} = 0 \quad \text{vs} \quad H1: \text{markup} > 0$

- We reject the null hypothesis at the 1% threshold for all categories: Walkenhorst applies a statistically significant markup on all the categories of products.
- The estimated average markup is 71.49%, with a very high t-statistic.

Note: we applied a Benjamini–Hochberg correction, adjusting p-values to control the false discovery rate, limiting the expected proportion of false positives among results declared statistically significant when testing multiple hypotheses.

Price per ounce test results

H0: Walkenhorst price = Walmart price vs H1: Walkenhorst price > Walmart price

Table 2: Unit Price Difference (Walmart - Walkenhorst) Test Results

Category	Mean Unit Price Difference	Mean Unit Price Walkenhorst	Mean Unit Price Walmart	Degrees of Freedom	t-statistic	P-Value
SEAFOOD	-0.47***	0.91	0.4	112.35	8.15	0.00
MEATS	-0.44***	1.33	0.9	61.35	3.06	0.00
DIET, NUTRITION & SUPPLEMENTS	-0.38***	1.45	1.1	101.93	2.68	0.01
SNACKS	-0.37***	0.86	0.5	395.37	12.75	0.00
NUTS	-0.27***	0.89	0.6	91.42	3.72	0.00
SPICES & SEASONINGS	-0.27**	1.20	0.9	151.13	2.40	0.02
CEREAL & OATMEAL	-0.26***	0.68	0.4	66.39	3.77	0.00
GRANOLA BARS & BREAKFAST BARS	-0.26***	0.70	0.4	113.63	6.06	0.00
CRACKERS	-0.22***	0.66	0.4	65.54	3.37	0.00
All Products	-0.20***	0.77	0.6	2795.63	10.25	0.00
CONDIMENTS, SAUCES & SPREADS	-0.17***	0.53	0.4	187.80	3.44	0.00
COOKIES	-0.17***	0.60	0.4	194.08	4.88	0.00
MEALS & SIDES	-0.13***	0.54	0.4	390.37	4.41	0.00
CANDY	-0.03	0.92	0.9	257.80	0.50	0.64
DRINKS & BEVERAGES	-0.03*	0.11	0.1	34.24	1.84	0.08
PASTRIES	0.00	0.53	0.5	80.22	-0.04	0.97

*Note: * p<0.1; ** p<0.05; *** p<0.01.*

We reject the null hypothesis for all categories at the 1% threshold: Walkenhorst prices are statistically significantly higher than Walmart prices for all the categories of products. The estimated average price difference is 20 cents/ounce.

Regression results

Table 3: Regression Results-Log of Price per Ounce

Variable	Coefficient	Std. Error	t-statistic	p-Value
(Intercept)	-0.61***	(2.71e-02)	-2.26e+01	5.23e-104
Vendor	0.37***	(5.42e-02)	6.77e+00	1.58e-11
BREAD	-0.69***	(1.05e-14)	-6.54e+13	0.00e+00
CANDY	0.16***	(1.08e-14)	1.47e+13	0.00e+00
CEREAL & OATMEAL	-0.35***	(1.06e-14)	-3.35e+13	0.00e+00
CHEESE	-0.45***	(1.05e-14)	-4.29e+13	0.00e+00
COFFEE	0.45***	(1.05e-14)	4.25e+13	0.00e+00
CONDIMENTS, SAUCES & SPREAD	-0.62***	(1.13e-14)	-5.43e+13	0.00e+00
COOKIES	-0.40***	(1.05e-14)	-3.83e+13	0.00e+00
CRACKERS	-0.28***	(1.05e-14)	-2.66e+13	0.00e+00
CREAMERS	-0.62***	(1.05e-14)	-5.91e+13	0.00e+00
DIET, NUTRITION & SUPPLEMENT	0.47***	(1.05e-14)	4.48e+13	0.00e+00
DRINK MIXES	0.15***	(1.07e-14)	1.43e+13	0.00e+00
DRINKS & BEVERAGES	-2.05***	(1.06e-14)	-1.93e+14	0.00e+00
FRUITS & VEGETABLES	-1.07***	(1.04e-14)	-1.02e+14	0.00e+00
GRANOLA BARS & BREAKFAST BAKED	-0.22***	(1.06e-14)	-2.09e+13	0.00e+00
MEALS & SIDES	-0.50***	(1.05e-14)	-4.71e+13	0.00e+00
MEATS	0.35***	(1.05e-14)	3.32e+13	0.00e+00
NUTS	0.01***	(1.06e-14)	8.10e+11	0.00e+00
OLIVES, PICKLES & PEPPERS	-0.95***	(1.07e-14)	-8.87e+13	0.00e+00
PASTRIES	-0.41***	(1.05e-14)	-3.88e+13	0.00e+00
SEAFOOD	-0.13***	(1.07e-14)	-1.23e+13	0.00e+00
SNACKS	-0.09***	(1.00e-14)	-9.32e+12	0.00e+00
SPICES & SEASONINGS	0.30***	(1.06e-14)	2.83e+13	0.00e+00
SWEETENERS	-0.19***	(1.04e-14)	-1.79e+13	0.00e+00
TACO SHELLS & TORTILLAS	-0.63***	(1.05e-14)	-6.00e+13	0.00e+00

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Walmart is the reference vendor (vendor = 1 for Walkenhorst, 0 for Walmart).
Asian Food is the reference category.
Observations: $n=2800$.
Standard Errors are clustered at the category level using the Bell-McCaffrey (2002) bias-reduced correction.

Controlling for category, Walkenhorst products are 37% higher than Walmart’s. This result is statistically significant at the 1% threshold.

Display of possible extensions and limitations

Discussion of the study limitations

Economies of scale bias

Walmart often sells products in large bulk packs, whereas Walkenhorst tends to sell smaller individual portions. Even when comparing price-per-ounce, this does not account for the natural "economy of scale" price reduction found in larger retail packages.

The absence of bulk quantity options in the Walkenhorst catalog could represent a (monopolistic) choice constraint in and of itself.

One potential (partial) solution we considered was to **control for product weight** in the regression. However, this would have required extracting weight information from Walmart observations which appear as character strings with heterogeneous and inconsistent patterns. Due to time constraints, we were unable to reliably construct this quantity measure, and therefore did not include it in the analysis.

A negligible number of outliers were present in the Walmart data. We excluded these observations to avoid biasing our estimates; however, some of them may not be true outliers but rather reflect price outcomes driven by economies of scale.

Lack of other control variables

We initially intended to control for the nutritional score and brand effects. However, nutritional score information was unavailable for the Walkenhorst dataset. In addition, brand identification in the Walmart data proved difficult, as brand names appear in product names or description as character strings with inconsistent and heterogeneous formats. As a result, we were unable to reliably construct these control variables and did not include them in the analysis.

Extensions and directions for future research

Basket of goods analysis

Assuming we are able to gather information on a typical basket of goods, a **detailed analysis of the total markup** could be performed. Allowing to provide a more **accurate assessment of the consumer loss** as well as of the economic impact on incarcerated people and families.

Going from price comparison to welfare analysis

Geographic extension for external validity

Future research could expand the analysis beyond California and Walkenhorst to gain external validity of the observed pricing patterns, and **potentially identify systematic pricing differences**. A country wide study could **support further regulatory recommendation to effectively regulate this activity**.

With extensive information:

- **Demand elasticity evaluation** across product categories, testing whether necessity goods exhibit higher markups **even though basic needs are covered in prisons**.
- **Linking observed prices to contract characteristics** (exclusivity clauses, commission rates) to identify how **revenue-sharing arrangements** distort pricing.

Appendix

List of references

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